

Isameldin Ahmed

Mechatronics Engineering Graduate

eldeenisam@gmail.com | +601139009572 | Kuala Lumpur, Malaysia | [linkedin.com/in/isameldin](https://www.linkedin.com/in/isameldin)

isameldin.netlify.app

SUMMARY

I am a recent Mechatronics Engineering graduate with a lifelong passion for engineering not just as a career, but as a way of life. Building solutions is what drew me to this degree in the first place, and it's the same instinct I bring to everyday problems, whether that's a project at work or something in my own home. In my free time, I build robotics projects and find ways to automate everyday tasks using the skills I've learned as well as always looking for the next thing to fix, build or improve.

WORK EXPERIENCE

Asia Pacific University

Senior Engineering Technical Assistant (Part-Time)

12/2022 – Present | Kuala Lumpur, Malaysia

- Promoted to lead a team of multiple lab assistants, while continuing to support 20-50 students per semester across engineering lab sessions.
- Guided students hands-on with equipment such as 3D printers, oscilloscopes, robotic arms, PLC units, as well as digital and analogue signal training kits.

Engineering Technical Assistant (Part-Time)

12/2024 – 12/2024 | Kuala Lumpur, Malaysia

- Maintained and repaired lab equipment including 3D printers, oscilloscopes, robotic arms, PLC units, and Arduino kits for electrical, mechanical, and mechatronics courses.
- Supported 30-40 students per semester during lab sessions, teaching hands on skills such as sensor calibration and analogue circuit design.
- Followed safety procedures and checked equipment regularly to keep the lab accident free.

Ambani Consortium Group (M) Sdn. Bhd.

03/2025 – 09/2025 | Kuala Lumpur, Malaysia

Mechatronics Engineering Intern

- Designed brackets and helped draft technical drawings for generator housings, working alongside a senior engineer.
- Helped connect generators to the internet by testing an LTE module that let users monitor and control their generators remotely through a web portal including turning them on or off from anywhere.
- Served on the project management team for the 2025 Invest Zambia International Conference, handling event planning and logistics.
- Traveled to Zambia for the conference, where I personally spoke with potential clients and government officials about the company's generators and a proposed water sanitation system.
- Learned the basics of ISO standards (9001, 14001, 45001) as they applied to the company's product development process.

EDUCATION

Bachelor of Mechatronic Engineering with Honours

06/2022 – 07/2026 | Kuala Lumpur, Malaysia

Asia Pacific University

Dual DMU Degree Programme. Second Class Upper Division.

Master of Engineering in Mechatronics Engineering, De Montfort University (DMU) Leicester

06/2022 – 07/2026

VOLUNTEERING & COMMUNITY LEADERSHIP

Head of Public Relations, Institution of Mechanical Engineers (IMechE) APU Student Section

08/2024 – 05/2026

Lead IMechE APU's communications and built industry partnerships — inviting speakers, doing public speaking at events, and helping organize over 20 events throughout my tenure.

TECHNICAL PROJECTS

EMG-Controlled Prosthetic Arm (Final Year Project)

Adapted the open-source HACKberry prosthetic arm design, 3D-printing and modifying it to add muscle-signal (EMG) control for hand movement, powered by an ESP32 microcontroller. Added temperature and pressure sensors so the user can feel through vibration feedback, with the arm automatically reacting to protect itself if temperatures get too high. Built an online dashboard connected to the ESP32 for calibrating the arm's sensors and controlling it remotely. Tested and improved the arm's muscle-signal control for reliable, accurate movement.

Automated Guided Vehicle (AGV) for Medical Equipment Transport in Hospitals (Group Design Project)

Contributed to a team built autonomous ground vehicle designed to transport medical equipment between hospitals. Worked on navigation using LiDAR, GPS, and camera based machine vision, running on ROS with an ESP32 and an NVIDIA Jetson Orin Nano. Designed the vehicle's chassis and sensor mounts in SolidWorks. Helped integrate different team members work such as sensor systems talking to the navigation logic.

Industrial Automation (Programmable Logic Controller)

Contributed to two independent PLC-controlled systems, mixing hands-on hardware work with programming, on a combination of real physical PLC hardware and simulation. One system moved objects between two fixed positions on a conveyor. The other sorted items by size and stamped them accordingly, using ladder logic to control the sequence.

CERTIFICATES

- MathWorks: MATLAB Onramp, Simulink Onramp, Control Design Onramp with Simulink
- Cisco Networking Academy: Introduction to IoT
- FreeCodeCamp: Python, C
- Council for Six Sigma Certification: Lean Six Sigma White Belt

LANGUAGES

English

Native

Arabic

Bilingual

AWARDS

High School Valedictorian, *Licef Secondary School, Lusaka, Zambia*

Al-Khayyam Mathematics Award, *Licef Secondary School, Lusaka, Zambia*

SKILLS

Programming & Embedded Systems

C, Python, MATLAB, Arduino

Automation & Controls

PLC Programming, Sensor Integration & Calibration

Circuit Design & CAD

Digital Circuit Design, SOLIDWORKS, 3D Printing

INTERESTS

Electronics Tinkering, Building and improving Personal Gadgets & Automation Projects, English Literature, Football

REFERENCES

Prathap Appakutty, *Engineering Lab Team Manager*, Asia Pacific University
prathap@apu.edu.my, +60 12-697 4210